Foundations of Machine Learning (ST510)

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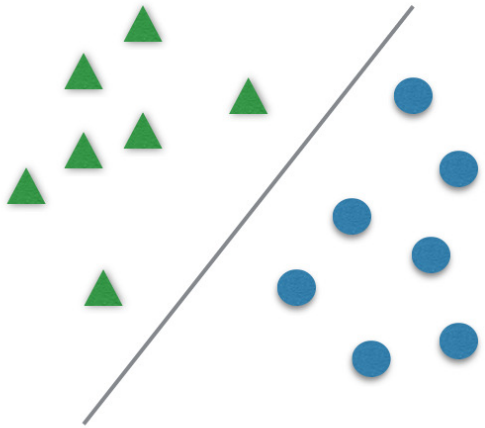
LSE

Motivating examples

Example-1: non-linear (large-margin) classification

- Given: $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$, $y_i \in \{-1, 1\}$.
- Goal: find an f classifier such that $f(\mathbf{x}) \approx y$.



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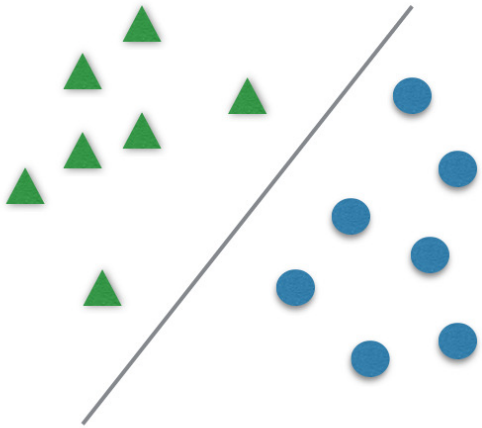

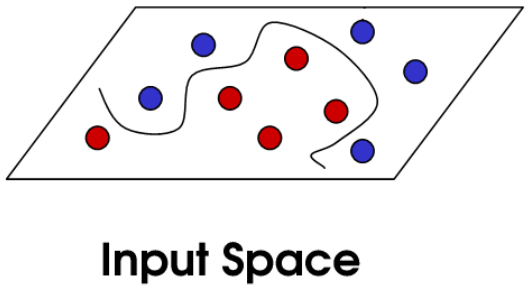

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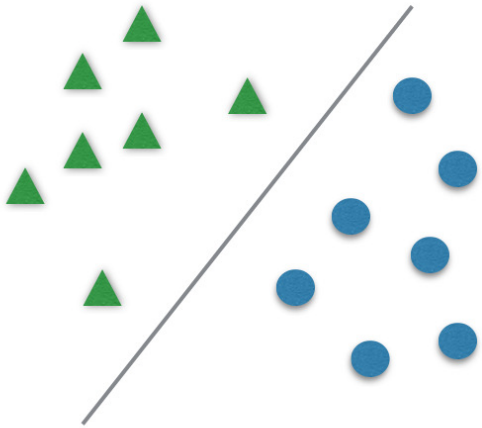
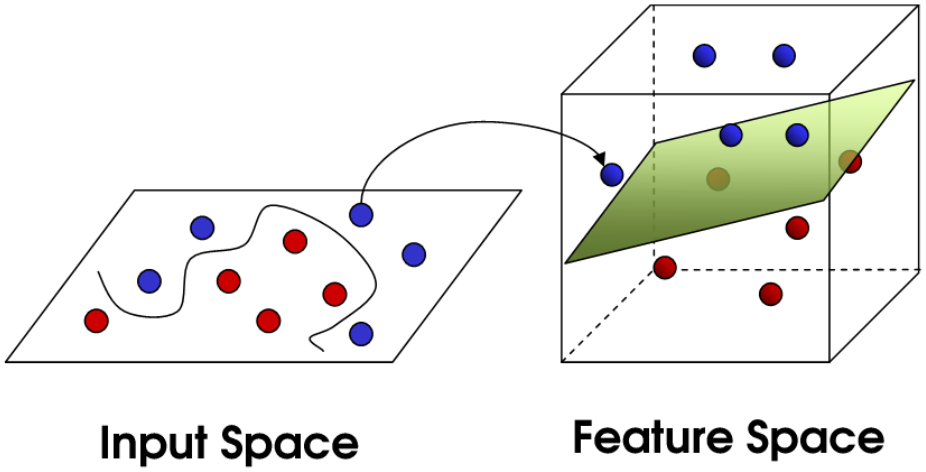
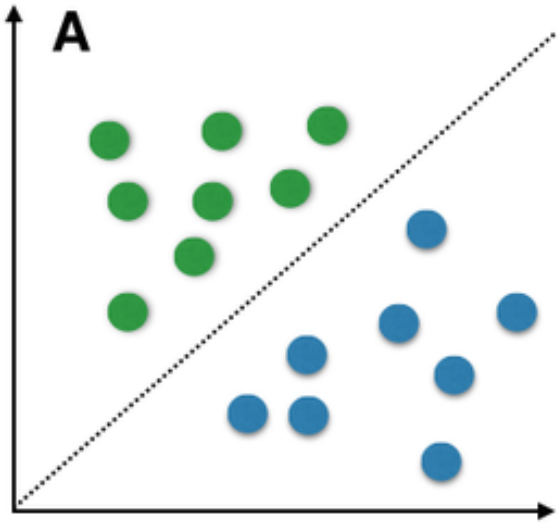




Example-1: continued – linear separability

Idealized situation



Decision surface:

$$\{\mathbf{x} : \langle \mathbf{w}, \mathbf{x} \rangle = 0\}$$

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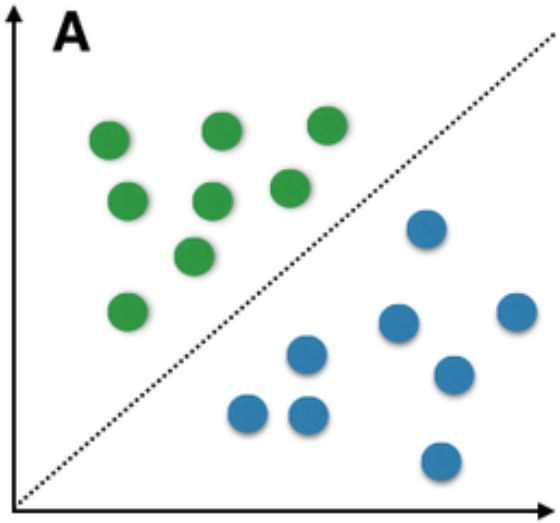

Decision surface:

$$\{\mathbf{x} : \langle \mathbf{w}, \mathbf{x} \rangle = 0\} \Rightarrow$$

classes:

$$\{\mathbf{x} : \langle \mathbf{w}, \mathbf{x} \rangle \geq 0\} \qquad \{\mathbf{x} : \langle \mathbf{w}, \mathbf{x} \rangle < 0\}$$

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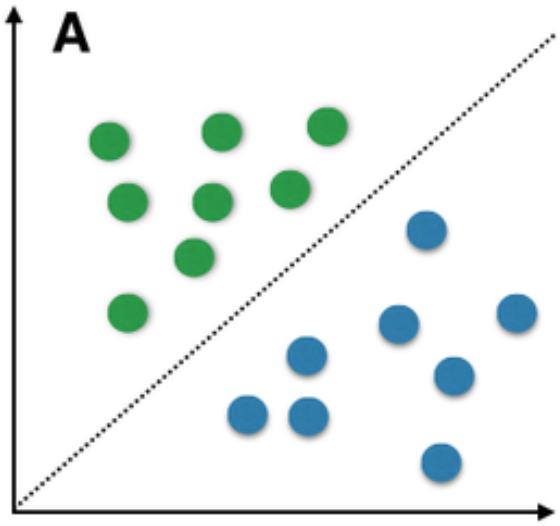
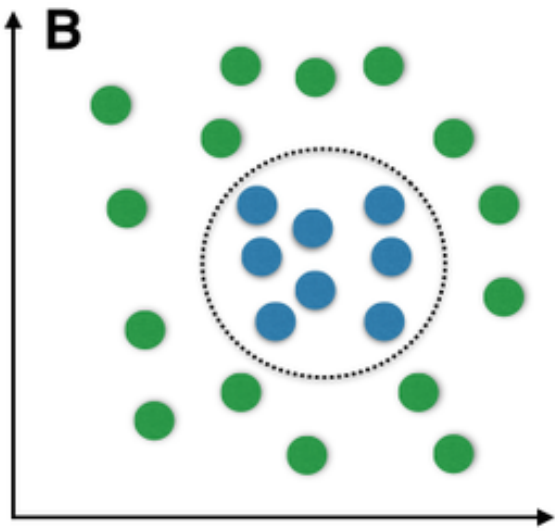


Real world



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Example-1: non-linear separability – continued

On the ellipse

$$\left\{ \mathbf{x} : \frac{(x_1 - c_1)^2}{a^2} + \frac{(x_2 - c_2)^2}{b^2} = 1 \right\}$$

Example-1: non-linear separability – continued

On the ellipse, outside

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On the ellipse, outside, inside:

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With polynomial feature: $\varphi(\mathbf{x}) = (x_1^2, x_1, 1, x_2^2, x_2)$:

- Decision surface: $\{\mathbf{x} : \langle \mathbf{w}, \varphi(\mathbf{x}) \rangle = 0\}$.

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- Classes: $\{\mathbf{x} : \langle \mathbf{w}, \varphi(\mathbf{x}) \rangle > 0\}$, $\{\mathbf{x} : \langle \mathbf{w}, \varphi(\mathbf{x}) \rangle < 0\}$.

Example-1: quadratic & polynomial features

Still in $\mathbb{R}^2$:

$$\varphi(\mathbf{x}) = \left(x_1^2, \sqrt{2}x_1x_2, x_2^2\right),$$

$$\langle \varphi(\mathbf{x}), \varphi(\mathbf{x}') \rangle = ?$$

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$\langle \mathbf{x}, \mathbf{x}' \rangle^d = \langle \varphi(\mathbf{x}), \varphi(\mathbf{x}') \rangle$: $\varphi(\mathbf{x}) =$ d-order polynomial. $\Rightarrow$

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$\langle \mathbf{x}, \mathbf{x}' \rangle^d = \langle \varphi(\mathbf{x}), \varphi(\mathbf{x}') \rangle$: $\varphi(\mathbf{x}) =$ d-order polynomial. $\Rightarrow$ Explicit computation would be heavy!

Example-2: characterizing distributions / independence

- Given: random variable $(X, Y) \in \mathcal{X} \times \mathcal{Y}$, $(X, Y) \sim \mathbb{P}_{XY}$.
- Goal: to measure the dependence of X and Y.

Example-2: characterizing distributions / independence

- Given: random variable $(X, Y) \in \mathcal{X} \times \mathcal{Y}$, $(X, Y) \sim \mathbb{P}_{XY}$.
- Goal: to measure the dependence of X and Y.
- Desiderata for a $Q(\mathbb{P}_{XY})$ independence measure [**?**]:
  1. $Q(\mathbb{P}_{XY})$ is well-defined,
  2. $Q(\mathbb{P}_{XY}) \in [0, 1]$,
  3. $Q(\mathbb{P}_{XY}) = 0$ iff. $X \perp Y$.
  4. $Q(\mathbb{P}_{XY}) = 1$ iff. $Y = f(X)$ or $X = g(Y)$.

Example-2: continued

- He showed:

$$Q(\mathbb{P}_{XY}) = \sup_{f,g:\ \text{measurable}} \operatorname{corr}(f(X), g(Y)),$$

  satisfies 1-4.
- Too ambitious:
  - computationally intractable.
  - many measurable functions.

Example-2: continued; measurable $\rightarrow$ continuous

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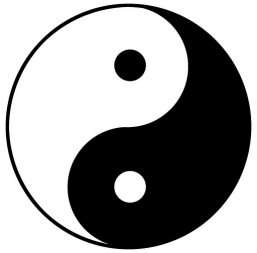
Key: balance

denseness $\to$ universality, computation $\to$ RKHS.



Motivation: kernels = generalized inner product

1. Various data types.
2. RKHS: flexible ($\overset{1:1}{\longleftrightarrow}$ probability measures).
3. Still computationally tractable: enough $k(x_i, x_j) \in \mathbb{R}$.
4. RKHS: Hilbert $\Rightarrow$ statistical analysis.
5. v-RKHS $[k(x, x') \in \mathcal{L}(Y)]$: dependency among output coordinates.

Kernel, RKHS: definition, kernel factory

Kernel, RKHS : generalized inner product, -linear methods

- Def-1 (feature space):

$$k(x, y) = \langle \varphi(x), \varphi(y) \rangle_{\mathcal{H}} \quad x, y \in \mathcal{X}.$$

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$$k(x, y) = \langle \varphi(x), \varphi(y) \rangle_{\mathcal{H}} \quad x, y \in \mathcal{X}.$$

- Def-2 (reproducing kernel):

$$k(\cdot, x) \in \mathcal{H}, \qquad f(x) = \langle f, k(\cdot, x) \rangle_{\mathcal{H}}.$$

Constructively, $\mathcal{H}_k = \overline{\{\sum_{i=1}^{n} \alpha_i k(\cdot, x_i)\}}$.

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- Def-3 (Gram matrix): $\mathbf{G} = [k(x_i, x_j)]_{i,j=1}^{n} \in \mathbb{R}^{n \times n} \succeq 0$.
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- All these definitions are equivalent, $k \overset{1:1}{\leftrightarrow} \mathcal{H}_k$.
- Examples on $\mathbb{R}^d$ ($\gamma > 0$, $p \in \mathbb{Z}^+$): $k_p(\mathbf{x}, \mathbf{y}) = (\langle \mathbf{x}, \mathbf{y} \rangle + \gamma)^p$, $k_G(\mathbf{x}, \mathbf{y}) = e^{-\gamma \|\mathbf{x} - \mathbf{y}\|_2^2}$, $k_e(\mathbf{x}, \mathbf{y}) = e^{-\gamma \|\mathbf{x} - \mathbf{y}\|_2}$.

Some kernel-enriched domains: $(\mathcal{X}, k)$

- Strings [?, ?, ?, ?, ?, ?, ?],
- time series [?, ?, ?, ?],
- trees [?, ?],
- groups and specifically rankings [?, ?],
- sets [?, ?, ?, ?], probability distributions [?, ?, ?, ?],
- various generative models [?, ?, ?, ?],
- fuzzy domains [?], or
- graphs [?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?].

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  2. Cone. If $k_m : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ kernel, $\alpha_m \geq 0$ ($m = 1, \ldots, M$), then

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&= \langle \varphi(x), \varphi(y) \rangle_{\mathcal{H}} , \\
\varphi(x) &= (\sqrt{\alpha_1}\varphi_1(x), \ldots, \sqrt{\alpha_M}\varphi_M(x)) \in \mathcal{H} := \oplus_{m=1}^{M} \mathcal{H}_m .
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Example: $\oplus_{m=1}^{M} \mathbb{R} = \mathbb{R}^M$.

Kernel factory – continued

4. Product. If $(k_m)_{m=1}^{M}$ are kernels on $\mathcal{X}_m$, then

$$\left(\otimes_{m=1}^{M} k_m\right)\left((x_1, \ldots, x_M), (x_1', \ldots, x_M')\right) = \prod_{m=1}^{M} k_m\left(x_m, x_m'\right).$$

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4. Product. If $(k_m)_{m=1}^M$ are kernels on $\mathcal{X}_m$, then

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- Thus, $(k_m)_{m=1}^M : \mathcal{X}\times\mathcal{X}\to\mathbb{R}$ kernels $\Rightarrow \prod_{m=1}^M k_m(x,x')$: kernel on $\mathcal{X}$.

Kernel factory – continued

4. Product. If $(k_m)_{m=1}^M$ are kernels on $\mathcal{X}_m$, then

$$\left(\otimes_{m=1}^M k_m\right)\left((x_1,\ldots,x_M),(x_1',\ldots,x_M')\right) = \prod_{m=1}^M k_m\left(x_m,x_m'\right).$$

- Thus, $(k_m)_{m=1}^M : \mathcal{X}\times\mathcal{X}\to\mathbb{R}$ kernels $\Rightarrow \prod_{m=1}^M k_m(x,x')$: kernel on $\mathcal{X}$.
- Consequence ($\gamma \geq 0$, $p\in\mathbb{Z}^+$):

$$k(\mathbf{x},\mathbf{y}) = \left(\langle \mathbf{x},\mathbf{y}\rangle_2 + \gamma\right)^p$$

is a kernel.

Kernel factory: product indeed

Let $M = 2$ and assume that $\varphi_m(x) \in \mathbb{R}^{d_m}$:

$$(k_1 \otimes k_2)\left((x, y), (x', y')\right) = k_1(x, x')k_2(y, y')$$

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Let $M = 2$ and assume that $\varphi_m(x) \in \mathbb{R}^{d_m}$:

$$\begin{aligned}(k_1 \otimes k_2)\left((x,y),(x',y')\right) &= k_1(x,x')k_2(y,y') \\ &= \left\langle \varphi_1(x), \varphi_1(x') \right\rangle_{\mathcal{H}_1} \left\langle \varphi_2(y), \varphi_2(y') \right\rangle_{\mathcal{H}_2}\end{aligned}$$

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&= \mathrm{tr}\left(\varphi_1(x)^\top \varphi_1(x') \varphi_2\left(y'\right)^\top \varphi_2(y)\right)
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&= \left\langle \underbrace{\varphi_1(x)\varphi_2(y)^\top}_{\in \mathbb{R}^{d_1 \times d_2}}, \underbrace{\varphi_1(x')\varphi_2(y')^\top}_{\in \mathbb{R}^{d_1 \times d_2}} \right\rangle_{\mathsf{F}},
\end{aligned}
$$

where $\langle \mathbf{A}, \mathbf{B} \rangle_{\mathsf{F}} = \mathrm{tr}\left(\mathbf{A}^\top \mathbf{B}\right) = \sqrt{\sum_{ij} A_{ij} B_{ij}}$ is the Frobenius inner product.

Kernel factory

6. Limit. If $(k_n)_{n\in\mathbb{N}}$ are kernels on $\mathcal{X}$, then

$$k(x, x') := \lim_{n\to\infty} k_n(x, x')$$

is a kernel. Why?

Kernel factory

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$$k(\mathbf{x}, \mathbf{y}) = e^{\gamma\langle\mathbf{x},\mathbf{y}\rangle_2} = \sum_{n\in\mathbb{N}} \frac{\left(\gamma\,\langle\mathbf{x}, \mathbf{y}\rangle_2\right)^n}{n!}$$

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Reason: polynomial kernel & limit rule.

Kernel factory – continued

7. Pre-post multiplication. k kernel on $\mathcal{X}$, $f : \mathcal{X} \to \mathbb{R}$, then

$$\tilde{k}(x, y) = f(x)k(x, y)f(y)$$

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Example (Gaussian kernel, $\gamma > 0$): previous example & new rule

$$k(\mathbf{x}, \mathbf{y}) = e^{-\gamma\|\mathbf{x}-\mathbf{y}\|_2^2}$$

by using $\|\mathbf{x} - \mathbf{y}\|_2^2 = \|\mathbf{x}\|_2^2 + \|\mathbf{y}\|_2^2 - 2\langle\mathbf{x}, \mathbf{y}\rangle$.

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$$k(\mathbf{x}, \mathbf{y}) = e^{-\gamma\|\mathbf{x}-\mathbf{y}\|_2^2} = e^{-\gamma\|\mathbf{x}\|_2^2} e^{2\gamma\langle\mathbf{x},\mathbf{y}\rangle_2} e^{-\gamma\|\mathbf{y}\|_2^2}$$

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Properties of $\mathcal{H}_k$, computational tractability

Properties of k control that of $\mathcal{H}_k$

[**?**, Chapter 4]:

- k: bounded $[\sup_{x,y \in \mathcal{X}} k(x,y) \leq C] \Rightarrow \forall f \in \mathcal{H}_k$ is bounded

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- k: analytic $\Rightarrow \forall f \in \mathcal{H}_k$ is analytic.

Representer theorem [?, ?]

- Given: $\{(x_i, y_i)\}_{i=1}^n$, say classification/regression.
- Goal:

$$J(f) = V(x_1, y_1, f(x_1), \ldots, x_n, y_n, f(x_n)) + r\left(\|f\|_{\mathcal{H}_k}^2\right) \to \min_{f \in \mathcal{H}_k},$$

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- Example:

$$V(\ldots) = \frac{1}{n}\sum_{i=1}^n \max(1 - y_i f(x_i), 0) \quad \text{(soft classification)},$$

$$V(\ldots) = \frac{1}{n}\sum_{i=1}^n [f(x_i) - y_i]^2 \qquad \text{(regression)}.$$

Representer theorem – continued

. . . then

- $\exists$ solution in the form:

$$f^* = \sum_{i=1}^{n} \alpha_i k(\cdot, x_i), \quad \alpha_i \in \mathbb{R}.$$

- r: strictly increasing $\Rightarrow \forall$ solution is of this form.
- Example: $r(z) = \lambda z$, $\lambda > 0$.

Representer theorem – proof

Objective

$$J(f) = V(x_1, y_1, f(x_1), \ldots, x_n, y_n, f(x_n)) + r\left(\|f\|^2_{\mathcal{H}_k}\right) \to \min_{f \in \mathcal{H}_k} .$$

Decompose & Pythagorean theorem:

$$\begin{aligned} S &= \operatorname{span}\left(k(\cdot, x_i) \,:\, i \in [n]\right), \\ f &= f_S + f_\perp, \\ \|f\|^2_{\mathcal{H}_k} &= \|f_S\|^2_{\mathcal{H}_k} + \underbrace{\|f_\perp\|^2_{\mathcal{H}_k}}_{\geq 0} \geq \|f_S\|^2_{\mathcal{H}_k} . \end{aligned}$$

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In J

- 1st term: depends on f_S only, $f(x_i) = \langle f, k(\cdot, x_i)\rangle_{\mathcal{H}_k}$.

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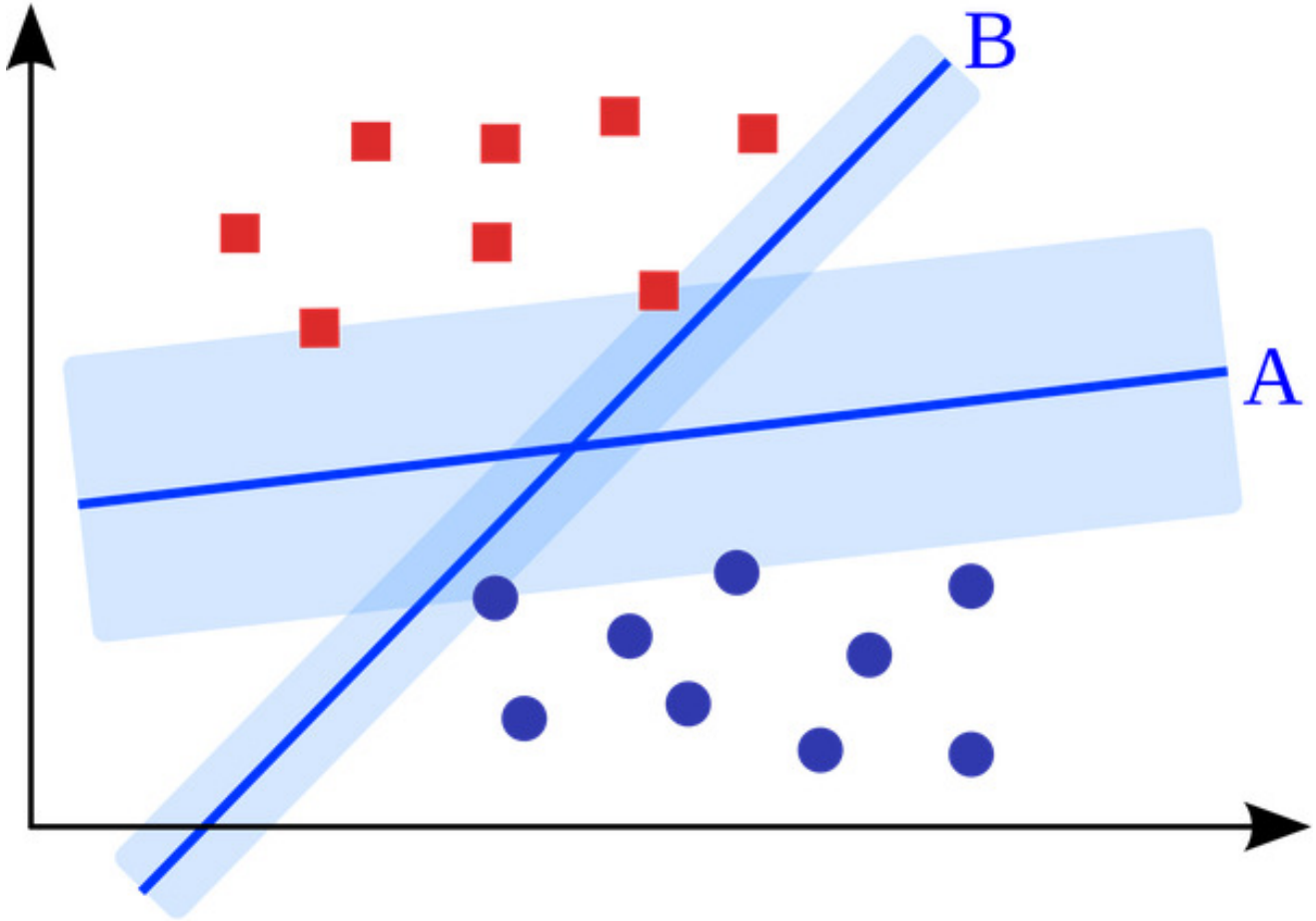
- 2nd term: can only decrease by neglecting $f_\perp$ ($r \nearrow$).

Classification: SVMC

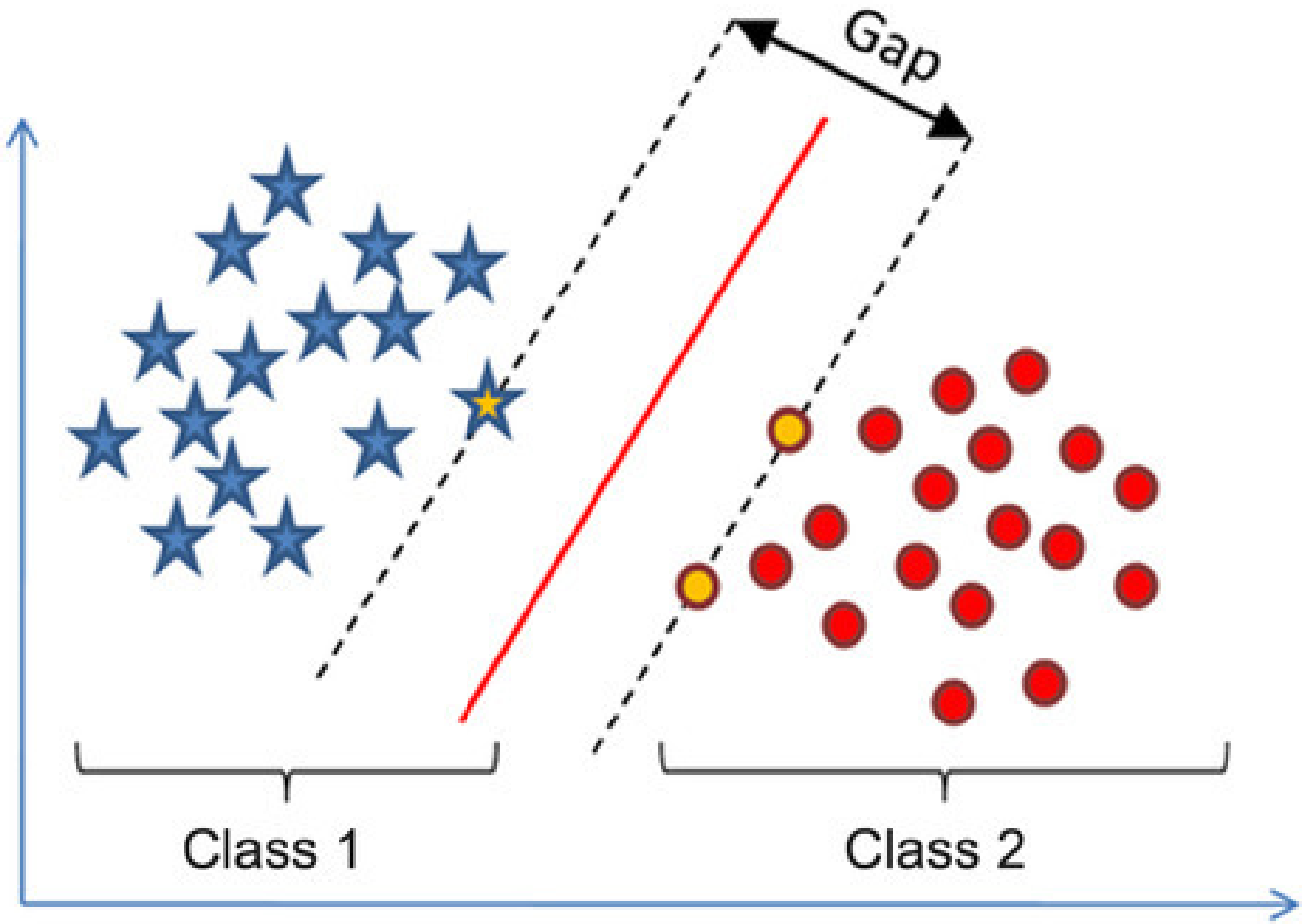
Support vector machine for classification: SVMC

Which separating line is the 'best'?



SVMC

Answer / intuition: the one with the largest margin.



SVM formulation: hard classification

- Hyperplane: $f_{\mathbf{w},b}(\mathbf{x}) = \langle \mathbf{w}, \mathbf{x} \rangle + b$,
  - $\mathbf{w}$: normal vector, b: offset.

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- Shortly,

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- Decision: $\hat{y}(\mathbf{x}) = \text{sign}\left(\langle \mathbf{w}, \mathbf{x} \rangle + b \right)$.

SVM formulation: soft classification

- Hard classification objective:

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There might not be solution! (non-linearly separable case)

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- Soft classification objective ($C > 0$):

$$\min_{\mathbf{w},b,\xi} \frac{1}{2} \|\mathbf{w}\|_2^2 + C \sum_{i=1}^{n} \xi_i \;\; \text{s.t.} \;\; y_i \left(\langle \mathbf{w}, \mathbf{x}_i \rangle + b\right) \geq 1 - \xi_i, \; \xi_i \geq 0, \; \forall i.$$

  Linear penalty on misclassification.

Note on the soft objective of SVMC

$$\min_{\mathbf{w},b,\xi} \frac{1}{2}\|\mathbf{w}\|_2^2 + C\sum_{i=1}^{n}\xi_i \text{ s.t. } y_i\left(\langle \mathbf{w}, \mathbf{x}_i\rangle + b\right) \geq 1 - \xi_i,\ \xi_i \geq 0,\ \forall i$$

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where $h(u) = \max(1-u, 0)$ is the hinge loss.

Note on the soft objective of SVMC – continued

The hinge loss is the convex envelope of the zero-one loss:

$$z(u) = \mathbb{I}_{\{u<0\}}, \qquad u = y_i f(x_i),$$
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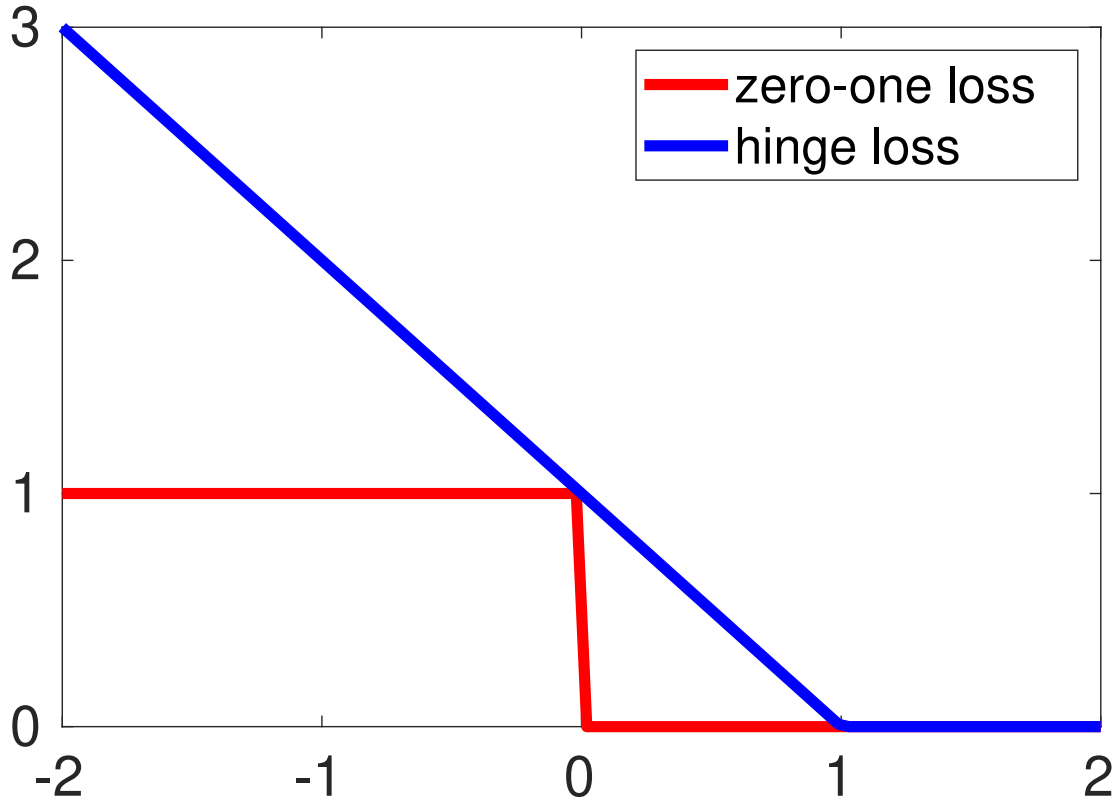

Soft classification – back to optimization

Soft classification objective:

$$\min_{\mathbf{w},b,\boldsymbol{\xi}} \frac{1}{2}\|\mathbf{w}\|_2^2 + C\sum_{i=1}^{n}\xi_i, \text{ s.t. } y_i\left(\langle \mathbf{w}, \mathbf{x}_i\rangle + b\right) \geq 1 - \xi_i,\ \xi_i \geq 0 \quad (\forall i).$$

Lagrangian function: with $\alpha_i \geq 0, \beta_i \geq 0$ $(\forall i)$

$$L(\mathbf{w}, b, \boldsymbol{\xi}; \boldsymbol{\alpha}, \boldsymbol{\beta}) = \text{objective - Lagrangian multipliers} \times \text{conditions}$$

$$= \frac{1}{2}\|\mathbf{w}\|_2^2 + C\sum_{i=1}^{n}\xi_i - \sum_{i=1}^{n}\alpha_i\left[y_i\left(\langle \mathbf{w}, \mathbf{x}_i\rangle + b\right) - 1 + \xi_i\right] - \sum_{i=1}^{n}\beta_i\xi_i.$$

Solving for $\frac{\partial L}{\partial \text{primal}} = 0$, we get ...

SVM formulation: soft classification

$$L(\mathbf{w}, b, \boldsymbol{\xi}; \boldsymbol{\alpha}, \boldsymbol{\beta}) =$$
$$= \frac{1}{2}\|\mathbf{w}\|_2^2 + C\sum_{i=1}^{n}\xi_i - \sum_{i=1}^{n}\alpha_i\left[y_i\left(\langle\mathbf{w}, \mathbf{x}_i\rangle + b\right) - 1 + \xi_i\right] - \sum_{i=1}^{n}\beta_i\xi_i.$$

Optimality equations:

$$\mathbf{0} = \frac{\partial L}{\partial \mathbf{w}} = \mathbf{w} - \sum_{i=1}^{n}\alpha_i y_i \mathbf{x}_i \quad (\mathbf{w} \leftrightarrow \boldsymbol{\alpha}),$$
$$0 = \frac{\partial L}{\partial b} = \sum_{i=1}^{n}\alpha_i y_i,$$
$$0 = \frac{\partial L}{\partial \xi_i} = C - \alpha_i - \beta_i.$$

Plugging these equations back to L, we have ...

SVM formulation: soft classification

Dual form:

$$\max_{\boldsymbol{\alpha}} \underbrace{\sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i,j=1}^{n} \alpha_i \alpha_j y_i y_j \langle \mathbf{x}_i, \mathbf{x}_j \rangle}_{\text{quadratic in } \boldsymbol{\alpha}}, \text{ s.t. } \underbrace{0 \leq \alpha_i \leq C, \sum_{i=1}^{n} \alpha_i y_i = 0}_{\text{linear in } \boldsymbol{\alpha}}.$$

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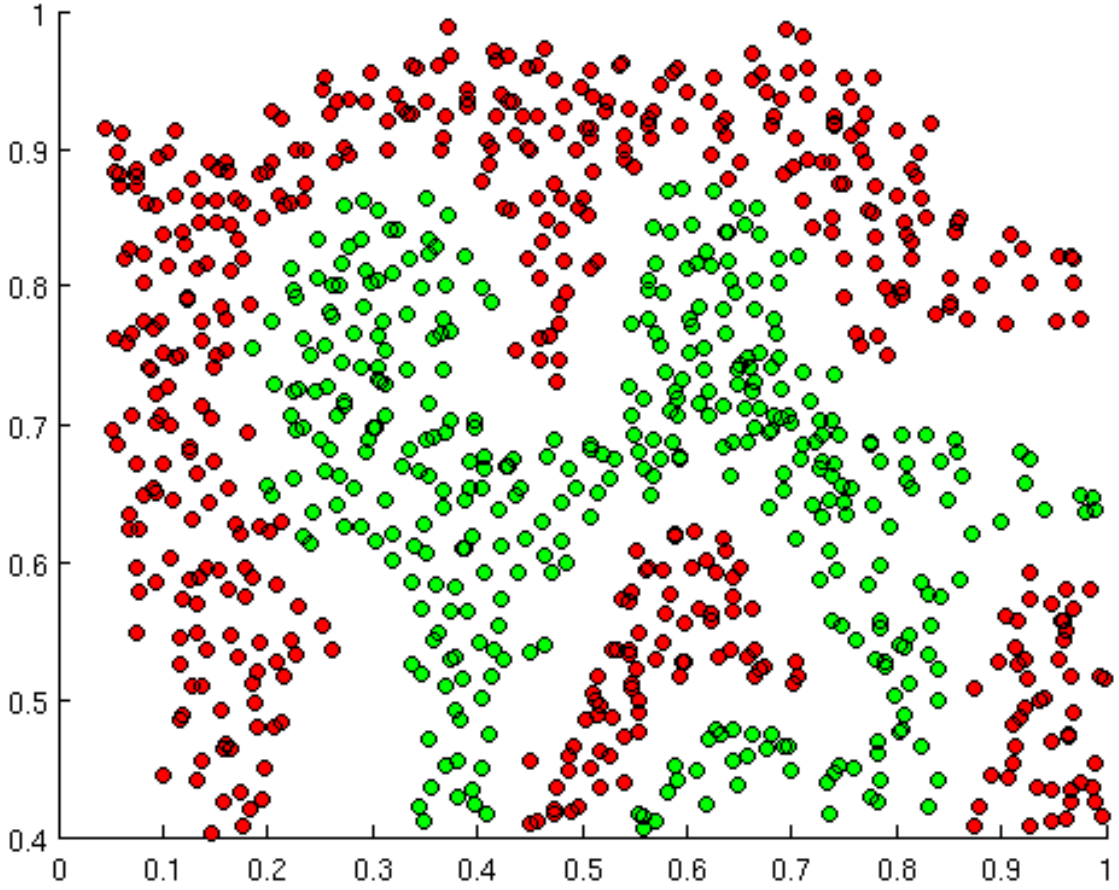
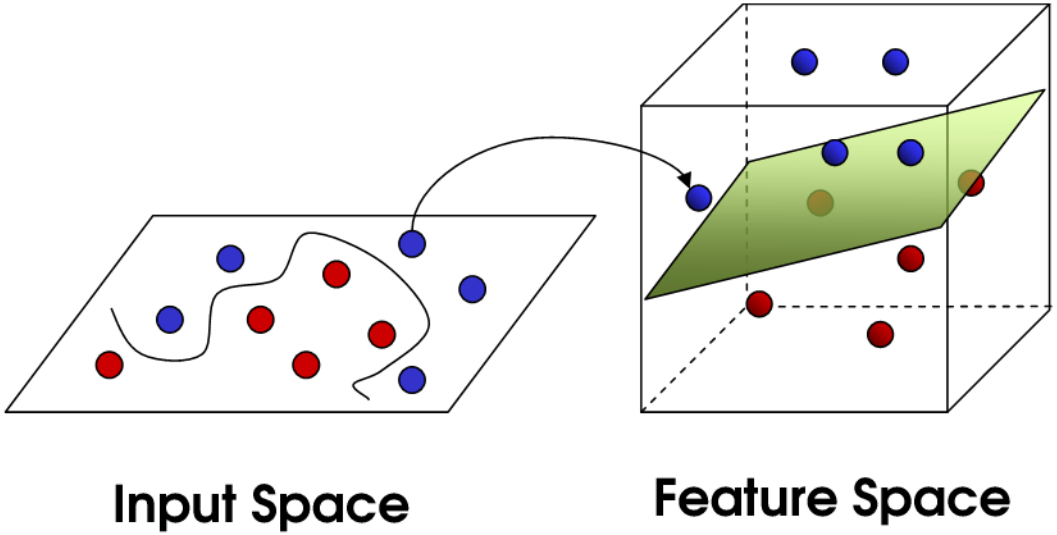
- Now:



If linear separability does not hold: kernel trick



Nonlinear SVM

- Linear SVM (dual):

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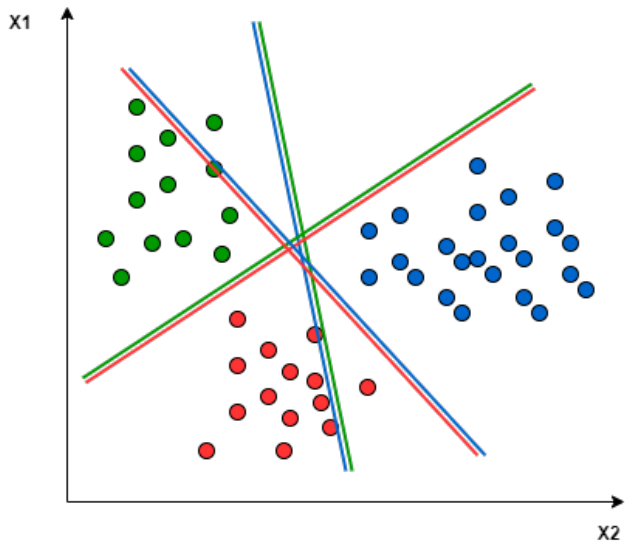
$$\min_{f \in \mathcal{H}_k, \boldsymbol{\xi}} \frac{1}{2} \|f\|_{\mathcal{H}_k}^2 + C \sum_{i=1}^{n} \xi_i, \text{ s.t. } y_i f(x_i) \geq 1 - \xi_i,\ \xi_i \geq 0,\ \forall i.$$

Multiclass (say M) classification with SVM

Idea

Break down the problem to multiple binary classification problems.

1. one-to-one approach:
   - $\frac{M(M-1)}{2}$ SVMC-s, i vs. j ($i \neq j$),
   - on new input x: the class with the most votes is predicted.



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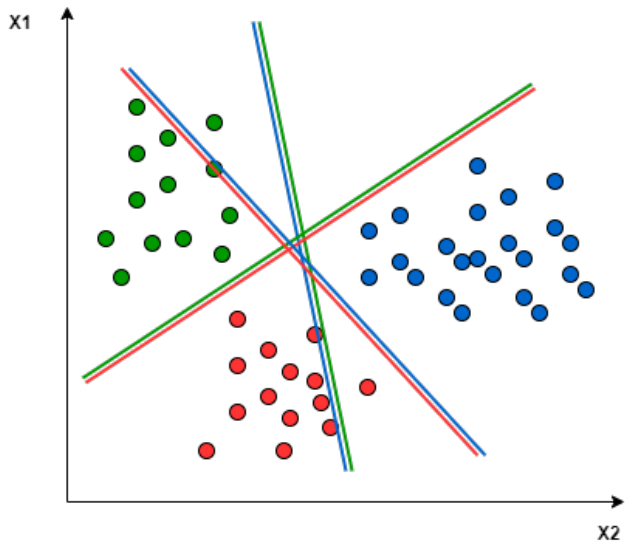
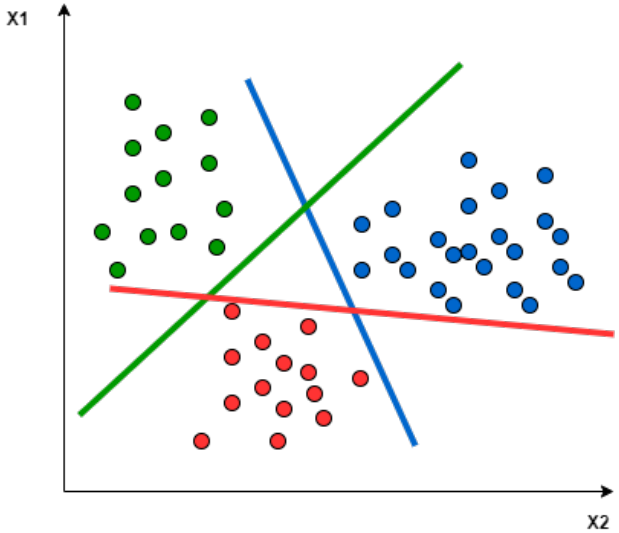
2. one-to-rest approach:
   - M SVMC-s, each predicts one class.
   - Classifiers give real-valued confidence scores: $f_m(x)$, $m \in [M]$.
   - Decision: $\hat{m} = \arg\max_{m \in [M]} f_m(x)$.





Parameter selection $\rightarrow$ cost: C, kernel parameter: σ

M-fold cross-validation $[\theta := (C, \sigma)]$:

1. Split data:
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   - test set: X_{te}, Y_{te}.

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4. Report: performance of θ^* on X_{te}, Y_{te}.

Regression: kernel ridge regression

Kernel ridge regression (KRR)

- Given: $\{(x_i, y_i)\}_{i=1}^n$, $\mathcal{H} := \mathcal{H}_k$, $y_i \in \mathbb{R}$.
- Task ($\lambda > 0$):

$$J(f) = \frac{1}{n} \sum_{i=1}^{n} [y_i - f(x_i)]^2 + \lambda \, \|f\|_{\mathcal{H}}^2 \to \min_{f \in \mathcal{H}} .$$

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- Analytical solution:

$$f(x) = [k(x_1, x), \ldots, k(x_n, x)] \left(\mathbf{G} + \lambda n \mathbf{I}_n\right)^{-1} [y_1; \ldots; y_n],$$
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Question

How do we get this solution?

Kernel ridge regression

By the representer theorem

$$f = \sum_{i=1}^{n} a_i k(\cdot, x_i), \quad (a_i \in \mathbb{R}).$$

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Multiplying the objective by n, using the reproducing property:

$$\begin{aligned}
\tilde{J}(f) &= \sum_{j=1}^{n} [y_j - \langle f, k(\cdot, x_j) \rangle_{\mathcal{H}}]^2 + \lambda n \, \|f\|_{\mathcal{H}}^2 \\
&= \|\mathbf{y} - \mathbf{G}\mathbf{a}\|_2^2 + (\lambda n) \mathbf{a}^\top \mathbf{G} \mathbf{a} \\
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Kernel machines: a simple algorithm = SGD [?]

Empirical regularized risk:

$$\min_{f \in \mathcal{H}_k} J(f) = \frac{1}{n} \sum_{i=1}^{n} \ell(f(x_i), y_i) + \frac{\lambda}{2} \|f\|_{\mathcal{H}_k}^2 \quad (\lambda > 0).$$

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Update (learning rate: $\eta_t > 0$):

$$f_{t+1} = f_t - \eta_t \underbrace{\left. \frac{\partial J_{\text{inst}}(f, (x_t, y_t))}{\partial f} \right|_{f=f_t}}_{\left. \frac{\partial \ell(z,y)}{\partial z} \right|_{z=f_t(x_t), y=y_t} k(\cdot, x_t) + \lambda f_t} .$$

Note: if ℓ is non-differentiable, subgradient is taken.

SGD implementation

1. Initialization: $f_1 = 0$.
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4. $\Leftrightarrow$ Update (in terms of coefficients): For $i \in [t]$,

$$\alpha_i := \begin{cases} -\eta_t \ell'(f_t(x_t), y_t) & \text{if } i = t, \\ (1 - \eta_t \lambda) \alpha_i & \text{if } i < t. \end{cases}$$

Optimizers+

1. Recall the dual problem:

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   - scaling to billions of points [**?**],
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3. For large-scale classification (+recent survey), see [**?**]:
   - Nyström technique + accelerated stochastic subgradient descent.

Maximal correlation: KCCA

KCCA: definition

- Given: $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$, $\ell : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}$.
- Associated:
  - feature maps $\varphi(x) = k(\cdot, x)$, $\psi(y) = \ell(\cdot, y)$,
  - RKHS-s $\mathcal{H}_k$, $\mathcal{H}_\ell$.

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  - RKHS-s $\mathcal{H}_k$, $\mathcal{H}_\ell$.
- KCCA measure of $(X, Y) \in \mathcal{X} \times \mathcal{Y}$

$$\rho_{\mathrm{KCCA}}(X, Y; \mathcal{H}_k, \mathcal{H}_\ell) = \sup_{f \in \mathcal{H}_k, g \in \mathcal{H}_\ell} \mathrm{corr}(f(X), g(Y)),$$

$$\mathrm{corr}(f(X), g(Y)) = \frac{\mathrm{cov}(f(X), g(Y))}{\sqrt{\mathrm{var}[f(X)]\,\mathrm{var}[g(Y)]}}.$$

KCCA: notes

- Optimization domain: $\mathcal{H}_k \times \mathcal{H}_\ell \ni (f, g)$.
- By reproducing property: we will get a finite-D task.
- k,ℓ linear: traditional CCA.
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Recall the reproducing property

$$f(x) = \langle f, k(\cdot, x) \rangle_{\mathcal{H}_k} \quad \forall f \in \mathcal{H}_k, x \in \mathcal{X}.$$

KCCA: empirical estimate

$$\widehat{\mathrm{cov}}(f(X), g(Y)) = \frac{1}{N}\sum_{n=1}^{N}\underbrace{\left[f(x_n) - \frac{1}{N}\sum_{i=1}^{N} f(x_i)\right]}_{\left\langle f, \varphi(x_n) - \frac{1}{N}\sum_{i=1}^{N}\varphi(x_i)\right\rangle_{\mathcal{H}_k}}\underbrace{\left[g(y_n) - \frac{1}{N}\sum_{i=1}^{N} g(y_i)\right]}_{\left\langle g, \psi(y_n) - \frac{1}{N}\sum_{i=1}^{N}\psi(y_i)\right\rangle_{\mathcal{H}_\ell}}$$

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Similarly:

$$\widehat{\mathrm{var}}[f(X)] = \frac{1}{N}\sum_{n=1}^{N} \left[f(x_n) - \frac{1}{N}\sum_{i=1}^{N} f(x_i) \right]^2$$

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$$\widehat{\mathrm{var}}[f(X)] = \frac{1}{N}\sum_{n=1}^{N}\left[f(x_n) - \frac{1}{N}\sum_{i=1}^{N} f(x_i)\right]^2 = \frac{1}{N}\sum_{n=1}^{N}\langle f, \tilde{\varphi}(x_n)\rangle_{\mathcal{H}_k}^2,$$

KCCA: empirical estimate

$$\widehat{\mathrm{cov}}(f(X), g(Y)) = \frac{1}{N}\sum_{n=1}^{N} \underbrace{\left[f(x_n) - \frac{1}{N}\sum_{i=1}^{N} f(x_i) \right]}_{\left\langle f, \varphi(x_n) - \frac{1}{N}\sum_{i=1}^{N}\varphi(x_i) \right\rangle_{\mathcal{H}_k}} \underbrace{\left[g(y_n) - \frac{1}{N}\sum_{i=1}^{N} g(y_i) \right]}_{\left\langle g, \psi(y_n) - \frac{1}{N}\sum_{i=1}^{N}\psi(y_i) \right\rangle_{\mathcal{H}_\ell}}$$

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$$\widehat{\mathrm{var}}[g(Y)] = \frac{1}{N}\sum_{n=1}^{N} \langle g, \tilde{\psi}(y_n)\rangle_{\mathcal{H}_\ell}^2.$$

KCCA: empirical estimate

- f: appears only as $\langle f, \tilde{\varphi}(x_n) \rangle_{\mathcal{H}_k}$ [similarly: g in $\langle g, \tilde{\psi}(y_n) \rangle_{\mathcal{H}_\ell}$]. $\Rightarrow$

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- $\forall$ component of f $\perp$

$$\operatorname{span}\left(\{\tilde{\varphi}(x_n)\}_{n=1}^N\right) = \left\{\sum_{n=1}^N c_n \tilde{\varphi}(x_n), \mathbf{c} = [c_n] \in \mathbb{R}^N\right\}$$

has no effect in the objective.

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has no effect in the objective.

Key idea

Enough to consider $f = \sum_{i=1}^N c_i \tilde{\varphi}(x_i)$, $g = \sum_{i=1}^N d_i \tilde{\psi}(y_i)$.

KCCA: empirical estimate

Using that $f = \sum_{i=1}^{N} c_i \tilde{\varphi}(x_i)$, $g = \sum_{i=1}^{N} d_i \tilde{\psi}(y_i)$:

$$\langle f, \tilde{\varphi}(x_n) \rangle_{\mathcal{H}_k} = \sum_{i=1}^{N} c_i \langle \tilde{\varphi}(x_i), \tilde{\varphi}(x_n) \rangle_{\mathcal{H}_k}$$

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$$\langle g, \tilde{\psi}(y_n) \rangle_{\mathcal{H}_\ell} = (\mathbf{d}^\top \tilde{\mathbf{G}}_Y)_n,$$

with the centered kernels ($\tilde{k}$, $\tilde{\ell}$) and Gram matrices ($\tilde{\mathbf{G}}_X$, $\tilde{\mathbf{G}}_Y$).

Until now

All the objective terms can be expressed by $\mathbf{c}$, $\mathbf{d}$, $\tilde{\mathbf{G}}_X$, $\tilde{\mathbf{G}}_Y$.

KCCA: empirical estimate

$$\widehat{\mathrm{cov}}(f(X), g(Y)) = \frac{1}{N}\sum_{n=1}^{N} \langle f, \tilde{\varphi}(x_n)\rangle_{\mathcal{H}_k} \langle g, \tilde{\psi}(y_n)\rangle_{\mathcal{H}_\ell},$$

$$\widehat{\mathrm{var}}[f(X)] = \frac{1}{N}\sum_{n=1}^{N} \langle f, \tilde{\varphi}(x_n)\rangle^2_{\mathcal{H}_k}, \widehat{\mathrm{var}}[g(Y)] = \frac{1}{N}\sum_{n=1}^{N} \langle g, \tilde{\psi}(y_n)\rangle^2_{\mathcal{H}_\ell},$$

and we have

$$\langle f, \tilde{\varphi}(x_n)\rangle_{\mathcal{H}_k} = (\mathbf{c}^\top \tilde{\mathbf{G}}_X)_n, \qquad \langle g, \tilde{\psi}(y_n)\rangle_{\mathcal{H}_\ell} = (\mathbf{d}^\top \tilde{\mathbf{G}}_Y)_n.$$

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Thus,

$$\widehat{\mathrm{cov}}(f(X), g(Y)) = \frac{1}{N}\mathbf{c}^\top \tilde{\mathbf{G}}_X \tilde{\mathbf{G}}_Y \mathbf{d},$$

$$\widehat{\mathrm{var}}[f(X)] = \frac{1}{N}\mathbf{c}^\top (\tilde{\mathbf{G}}_X)^2 \mathbf{c}, \quad \widehat{\mathrm{var}}[g(Y)] = \frac{1}{N}\mathbf{d}^\top (\tilde{\mathbf{G}}_Y)^2 \mathbf{d}.$$

KCCA: finite-D form

Empirical estimate of KCCA:

$$\widehat{\rho_{\mathrm{KCCA}}}^{\mathrm{temp}}(X, Y; \mathcal{H}_k, \mathcal{H}_\ell) = \sup_{\mathbf{c}\in\mathbb{R}^N, \mathbf{d}\in\mathbb{R}^N} \frac{\mathbf{c}^\top \tilde{\mathbf{G}}_X \tilde{\mathbf{G}}_Y \mathbf{d}}{\sqrt{\mathbf{c}^\top (\tilde{\mathbf{G}}_X)^2 \mathbf{c}} \sqrt{\mathbf{d}^\top (\tilde{\mathbf{G}}_Y)^2 \mathbf{d}}}.$$

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In practice ($\kappa > 0$):

$$\widehat{\rho_{\mathrm{KCCA}}}(X, Y) := \widehat{\rho_{\mathrm{KCCA}}}(X, Y; \mathcal{H}_k, \mathcal{H}_\ell, \kappa)$$
$$= \sup_{\mathbf{c}\in\mathbb{R}^N, \mathbf{d}\in\mathbb{R}^N} \frac{\mathbf{c}^\top \tilde{\mathbf{G}}_X \tilde{\mathbf{G}}_Y \mathbf{d}}{\sqrt{\mathbf{c}^\top \left(\tilde{\mathbf{G}}_X + \kappa \mathbf{I}_N\right)^2 \mathbf{c}}\sqrt{\mathbf{d}^\top \left(\tilde{\mathbf{G}}_Y + \kappa \mathbf{I}_N\right)^2 \mathbf{d}}}.$$

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Question

How do we solve it?

KCCA: solution

Stationary points of $\widehat{\rho_{\mathrm{KCCA}}}(X, Y)$:

$$\mathbf{0} = \frac{\partial \widehat{\rho_{\mathrm{KCCA}}}(X, Y)}{\partial \mathbf{c}}, \qquad \mathbf{0} = \frac{\partial \widehat{\rho_{\mathrm{KCCA}}}(X, Y)}{\partial \mathbf{d}},$$

which simplifies to

$$\tilde{\mathbf{G}}_X \tilde{\mathbf{G}}_Y \mathbf{d} = \frac{(\mathbf{c}^\top \tilde{\mathbf{G}}_X \tilde{\mathbf{G}}_Y \mathbf{d})(\tilde{\mathbf{G}}_X + \kappa \mathbf{I}_N)^2 \mathbf{c}}{\mathbf{c}^\top (\tilde{\mathbf{G}}_X + \kappa \mathbf{I}_N)^2 \mathbf{c}}, \; \tilde{\mathbf{G}}_Y \tilde{\mathbf{G}}_X \mathbf{c} = \frac{(\mathbf{d}^\top \tilde{\mathbf{G}}_Y \tilde{\mathbf{G}}_X \mathbf{c})(\tilde{\mathbf{G}}_Y + \kappa \mathbf{I}_N)^2 \mathbf{d}}{\mathbf{d}^\top (\tilde{\mathbf{G}}_Y + \kappa \mathbf{I}_N)^2 \mathbf{d}}$$

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Normalization:

- $(\mathbf{c}, \mathbf{d})$: solution $\Rightarrow$ $(a\mathbf{c}, b\mathbf{d})$: solution $a, b \in \mathbb{R} \backslash \{0\}$.
- denominators $:= 1$.

KCCA: final task

Find the maximal eigenvalue, $\lambda := \mathbf{c}^\top \tilde{\mathbf{G}}_X \tilde{\mathbf{G}}_Y \mathbf{d}$, of the generalized eigenvalue problem:

$$\begin{bmatrix} \mathbf{0} & \tilde{\mathbf{G}}_X \tilde{\mathbf{G}}_Y \\ \tilde{\mathbf{G}}_Y \tilde{\mathbf{G}}_X & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{c} \\ \mathbf{d} \end{bmatrix} = \mathbf{c}^\top \tilde{\mathbf{G}}_X \tilde{\mathbf{G}}_Y \mathbf{d} \begin{bmatrix} (\tilde{\mathbf{G}}_X + \kappa \mathbf{I}_N)^2 & \mathbf{0} \\ \mathbf{0} & (\tilde{\mathbf{G}}_Y + \kappa \mathbf{I}_N)^2 \end{bmatrix} \begin{bmatrix} \mathbf{c} \\ \mathbf{d} \end{bmatrix}$$

$$\mathbf{A}\mathbf{z} = \lambda \mathbf{B}\mathbf{z}.$$

Note: Python implementation in the ITE toolbox ($M \geq 2$, with acceleration).

Summary

- Kernel, RKHS: generalized inner product, - linear methods.
- Computational tractability: representer theorem.
- Classification: SVMC.
- Regression: kernel ridge regression.
- Maximal correlation: KCCA.

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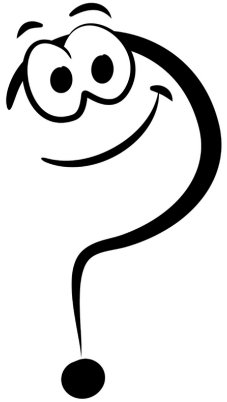

Contents (KCCA: questions)

1. Is KCCA an independence measure? ($\Leftarrow$ universality)
2. Meaning/handling of the regularization (κ).
3. $M \geq 2$ components.
4. Computation of $\tilde{\mathbf{G}}_X$, $\tilde{\mathbf{G}}_Y$.

Q1 (independence measure) $\Leftarrow$ universal k, ℓ

If $X \perp Y$, then $\rho_{\mathrm{KCCA}}(X, Y; \mathcal{H}_k, \mathcal{H}_\ell, \kappa) = 0$. Opposite direction:

- For 'rich' $\mathcal{H}_k$, $\mathcal{H}_\ell$ [**?**, **?**].

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- For 'rich' $\mathcal{H}_k$, $\mathcal{H}_\ell$ [**?**, **?**].
- Enough: universal kernel on a compact metric domain.
- Example ($\gamma > 0$):
  - Gaussian: $k(\mathbf{x}, \mathbf{x}') = e^{-\gamma \|\mathbf{x}-\mathbf{x}'\|_2^2}$.
  - Laplacian kernel: $k(\mathbf{x}, \mathbf{x}') = e^{-\gamma \|\mathbf{x}-\mathbf{x}'\|_2}$.

Q1: universal kernel, $C(\mathcal{X}) = \{f : \mathcal{X} \to \mathbb{R} \text{ continuous}\}$

Definition

Assume:

- $\mathcal{X}$: compact metric space.
- k: continuous kernel on $\mathcal{X}$.

k is called *(c)-universal* [**?**] if $\mathcal{H}_k$ is dense in $(C(\mathcal{X}), \|\cdot\|_\infty)$.

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- k: continuous, bounded $\Rightarrow \mathcal{H}_k \subset C(\mathcal{X})$ [**?**].
- Extensions of c-universality to non-compact spaces:
  - c_0-universality, cc-universality, ... [**?**, **?**, **?**].

Q1: properties of universal kernels [?, ?]

If k is universal, then

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  is a metric.
- The normalized kernel (recall: corr)

$$\tilde{k}(x,y) := \frac{k(x,y)}{\sqrt{k(x,x)k(y,y)}}$$

  is universal.

Q1: universal Taylor kernels [**?**, **?**]

- For an $C^\infty \ni f : (-r, r) \to \mathbb{R}$

$$f(t) = \sum_{n=0}^{\infty} a_n t^n \quad t \in (-r, r), \; r \in (0, \infty].$$

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$$f(t) = \sum_{n=0}^{\infty} a_n t^n \quad t \in (-r, r), \; r \in (0, \infty].$$

- If $a_n > 0 \; \forall n$, then

$$k(\mathbf{x}, \mathbf{y}) = f(\langle \mathbf{x}, \mathbf{y} \rangle)$$

is universal on $\mathcal{X} := \left\{ \mathbf{x} \in \mathbb{R}^d : \|\mathbf{x}\|_2 \leq \sqrt{r} \right\}$.

Q1: universal kernels on compact subsets of $\mathbb{R}^d$, $\alpha > 0$

- $k(\mathbf{x}, \mathbf{y}) = e^{\alpha \langle \mathbf{x}, \mathbf{y} \rangle}$: previous result with $f(t) = e^{\alpha t} \Rightarrow a_n = \frac{\alpha^n}{n!}$.

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- $k(\mathbf{x}, \mathbf{y}) = e^{-\alpha \|\mathbf{x} - \mathbf{y}\|_2^2}$: exp. kernel & normalization.

Q1: universal kernels on compact subsets of $\mathbb{R}^d$, $\alpha > 0$

- $k(\mathbf{x}, \mathbf{y}) = (1 - \langle \mathbf{x}, \mathbf{y} \rangle)^{-\alpha}$ binomial kernel
  - on $\mathcal{X}$ compact $\subset \{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x}\|_2 < 1\}$.
  - $f(t) = (1 - t)^{-\alpha} = \sum_{n=0}^{\infty} \underbrace{\binom{-\alpha}{n}(-1)^n}_{>0} t^n \quad (|t| < 1)$,

where $\binom{b}{n} = \sum_{i=1}^{n} \frac{b-i+1}{i}$.

Contents

Q2 (κ)

In fact, we estimated

$$\rho_{\mathsf{KCCA}}(X, Y; \mathcal{H}_k, \mathcal{H}_\ell, \kappa) = \sup_{f \in \mathcal{H}_k, g \in \mathcal{H}_\ell} \operatorname{corr}(f(X), g(Y); \kappa),$$

$$\operatorname{corr}(f(X), g(Y); \kappa) = \frac{\operatorname{cov}(f(X), g(Y))}{\sqrt{\operatorname{var}[f(X)] + \kappa \|f\|^2_{\mathcal{H}_k}} \sqrt{\operatorname{var}[g(Y)] + \kappa \|g\|^2_{\mathcal{H}_\ell}}}.$$

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For consistent KCCA estimate:

- $\kappa_N \to 0$ [**?**](spline-RKHS), [**?**] (general RKHS).
- analysis: covariance operators.

Contents

Q3 ($M \geq 2$): symmetry, other form

For

$$\begin{bmatrix} \mathbf{0} & \tilde{\mathbf{G}}_X \tilde{\mathbf{G}}_Y \\ \tilde{\mathbf{G}}_Y \tilde{\mathbf{G}}_X & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{c} \\ \mathbf{d} \end{bmatrix} = \mathbf{c}^\top \tilde{\mathbf{G}}_X \tilde{\mathbf{G}}_Y \mathbf{d} \begin{bmatrix} (\tilde{\mathbf{G}}_X + \kappa \mathbf{I}_N)^2 & \mathbf{0} \\ \mathbf{0} & (\tilde{\mathbf{G}}_Y + \kappa \mathbf{I}_N)^2 \end{bmatrix} \begin{bmatrix} \mathbf{c} \\ \mathbf{d} \end{bmatrix}$$

$([\mathbf{c}, \mathbf{d}], \lambda)$ solution $\Rightarrow$ $([-\mathbf{c}; \mathbf{d}], -\lambda)$: solution. Thus, eigenvalues:

$$\{\lambda_1, -\lambda_1, \ldots, \lambda_N, -\lambda_N\}.$$

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Adding the r.h.s. to both sides:

$$\begin{bmatrix} (\tilde{\mathbf{G}}_X + \kappa \mathbf{I}_N)^2 & \tilde{\mathbf{G}}_X\tilde{\mathbf{G}}_Y \\ \tilde{\mathbf{G}}_Y\tilde{\mathbf{G}}_X & (\tilde{\mathbf{G}}_Y + \kappa \mathbf{I}_N)^2 \end{bmatrix} \begin{bmatrix} \mathbf{c} \\ \mathbf{d} \end{bmatrix} = (1+\lambda) \begin{bmatrix} (\tilde{\mathbf{G}}_X + \kappa \mathbf{I}_N)^2 & \mathbf{0} \\ \mathbf{0} & (\tilde{\mathbf{G}}_Y + \kappa \mathbf{I}_N)^2 \end{bmatrix} \begin{bmatrix} \mathbf{c} \\ \mathbf{d} \end{bmatrix}$$

with eigenvalues $\{1+\lambda_1, 1-\lambda_1, \ldots, 1+\lambda_N, 1-\lambda_N\}$.

Q3 ($M \geq 2$)

2-variables $[(X, Y)]$:

$$\begin{bmatrix} (\tilde{\mathbf{G}}_X + \kappa \mathbf{I}_N)^2 & \tilde{\mathbf{G}}_X \tilde{\mathbf{G}}_Y \\ \tilde{\mathbf{G}}_Y \tilde{\mathbf{G}}_X & (\tilde{\mathbf{G}}_Y + \kappa \mathbf{I}_N)^2 \end{bmatrix} \begin{bmatrix} \mathbf{c} \\ \mathbf{d} \end{bmatrix} = (1 + \lambda) \begin{bmatrix} (\tilde{\mathbf{G}}_X + \kappa \mathbf{I}_N)^2 & \mathbf{0} \\ \mathbf{0} & (\tilde{\mathbf{G}}_Y + \kappa \mathbf{I}_N)^2 \end{bmatrix} \begin{bmatrix} \mathbf{c} \\ \mathbf{d} \end{bmatrix}$$

Q3 ($M \geq 2$)

2-variables $[(X, Y)]$:

$$\begin{bmatrix} (\tilde{\mathbf{G}}_X + \kappa \mathbf{I}_N)^2 & \tilde{\mathbf{G}}_X \tilde{\mathbf{G}}_Y \\ \tilde{\mathbf{G}}_Y \tilde{\mathbf{G}}_X & (\tilde{\mathbf{G}}_Y + \kappa \mathbf{I}_N)^2 \end{bmatrix} \begin{bmatrix} \mathbf{c} \\ \mathbf{d} \end{bmatrix} = (1+\lambda) \begin{bmatrix} (\tilde{\mathbf{G}}_X + \kappa \mathbf{I}_N)^2 & \mathbf{0} \\ \mathbf{0} & (\tilde{\mathbf{G}}_Y + \kappa \mathbf{I}_N)^2 \end{bmatrix} \begin{bmatrix} \mathbf{c} \\ \mathbf{d} \end{bmatrix}$$

For M-variables (pairwise dependence):

$$\begin{bmatrix} (\tilde{\mathbf{G}}_1 + \kappa \mathbf{I}_N)^2 & \tilde{\mathbf{G}}_1 \tilde{\mathbf{G}}_2 & \dots & \tilde{\mathbf{G}}_1 \tilde{\mathbf{G}}_M \\ \tilde{\mathbf{G}}_2 \tilde{\mathbf{G}}_1 & (\tilde{\mathbf{G}}_2 + \kappa \mathbf{I}_N)^2 & \dots & \tilde{\mathbf{G}}_2 \tilde{\mathbf{G}}_M \\ \vdots & \vdots & & \vdots \\ \tilde{\mathbf{G}}_M \tilde{\mathbf{G}}_1 & \tilde{\mathbf{G}}_M \tilde{\mathbf{G}}_2 & \dots & (\tilde{\mathbf{G}}_M + \kappa \mathbf{I}_N)^2 \end{bmatrix} \begin{bmatrix} \mathbf{c}_1 \\ \mathbf{c}_2 \\ \vdots \\ \mathbf{c}_M \end{bmatrix} =$$

$$\gamma \begin{bmatrix} (\tilde{\mathbf{G}}_1 + \kappa \mathbf{I}_N)^2 & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & (\tilde{\mathbf{G}}_2 + \kappa \mathbf{I}_N)^2 & \dots & \mathbf{0} \\ \vdots & \vdots & & \\ \mathbf{0} & \mathbf{0} & \dots & (\tilde{\mathbf{G}}_M + \kappa \mathbf{I}_N)^2 \end{bmatrix} \begin{bmatrix} \mathbf{c}_1 \\ \mathbf{c}_2 \\ \vdots \\ \mathbf{c}_M \end{bmatrix}.$$

Contents

Q4: Centered Gram matrix

In short

$\tilde{\mathbf{G}}_X = \mathbf{H}\mathbf{G}_X\mathbf{H}$ with $\mathbf{H} = \mathbf{I}_N - \frac{\mathbf{E}_N}{N}$; $\mathbf{H}; \mathbf{E}_N \in \mathbb{R}^{N \times N}$.

$$(\tilde{\mathbf{G}}_X)_{ij} = \tilde{k}(x_i, x_j) = \langle \tilde{\varphi}(x_i), \tilde{\varphi}(x_j) \rangle_{\mathcal{H}_k}$$

Q4: Centered Gram matrix

In short

$\tilde{\mathbf{G}}_X = \mathbf{H}\mathbf{G}_X\mathbf{H}$ with $\mathbf{H} = \mathbf{I}_N - \frac{\mathbf{E}_N}{N}$; $\mathbf{H}; \mathbf{E}_N \in \mathbb{R}^{N \times N}$.

$$
\begin{aligned}
(\tilde{\mathbf{G}}_X)_{ij} &= \tilde{k}(x_i, x_j) = \langle \tilde{\varphi}(x_i), \tilde{\varphi}(x_j) \rangle_{\mathcal{H}_k} \\
&= \langle \varphi(x_i) - \frac{1}{N}\sum_{n=1}^{N} \varphi(x_n), \varphi(x_j) - \frac{1}{N}\sum_{m=1}^{N} \varphi(x_m) \rangle_{\mathcal{H}_k}
\end{aligned}
$$

Q4: Centered Gram matrix

In short

$\tilde{\mathbf{G}}_X = \mathbf{H}\mathbf{G}_X\mathbf{H}$ with $\mathbf{H} = \mathbf{I}_N - \frac{\mathbf{E}_N}{N}$; $\mathbf{H}; \mathbf{E}_N \in \mathbb{R}^{N\times N}$.

$$
\begin{aligned}
(\tilde{\mathbf{G}}_X)_{ij} &= \tilde{k}(x_i, x_j) = \langle \tilde{\varphi}(x_i), \tilde{\varphi}(x_j) \rangle_{\mathcal{H}_k} \\
&= \langle \varphi(x_i) - \frac{1}{N}\sum_{n=1}^{N} \varphi(x_n), \varphi(x_j) - \frac{1}{N}\sum_{m=1}^{N} \varphi(x_m) \rangle_{\mathcal{H}_k} \\
&= (\mathbf{G}_X)_{ij} - \frac{1}{N}\sum_{m=1}^{N} (\mathbf{G}_X)_{im} - \frac{1}{N}\sum_{n=1}^{N} (\mathbf{G}_X)_{nj} - \frac{1}{N^2}\sum_{n,m=1}^{N} (\mathbf{G}_X)_{nm}
\end{aligned}
$$

Q4: Centered Gram matrix

In short

$\tilde{\mathbf{G}}_X = \mathbf{H}\mathbf{G}_X\mathbf{H}$ with $\mathbf{H} = \mathbf{I}_N - \frac{\mathbf{E}_N}{N}$; $\mathbf{H}; \mathbf{E}_N \in \mathbb{R}^{N\times N}$.

$$
\begin{aligned}
(\tilde{\mathbf{G}}_X)_{ij} &= \tilde{k}(x_i, x_j) = \langle \tilde{\varphi}(x_i), \tilde{\varphi}(x_j) \rangle_{\mathcal{H}_k} \\
&= \langle \varphi(x_i) - \frac{1}{N}\sum_{n=1}^{N} \varphi(x_n), \varphi(x_j) - \frac{1}{N}\sum_{m=1}^{N} \varphi(x_m) \rangle_{\mathcal{H}_k} \\
&= (\mathbf{G}_X)_{ij} - \frac{1}{N}\sum_{m=1}^{N} (\mathbf{G}_X)_{im} - \frac{1}{N}\sum_{n=1}^{N} (\mathbf{G}_X)_{nj} - \frac{1}{N^2}\sum_{n,m=1}^{N} (\mathbf{G}_X)_{nm} \\
&= \left(\mathbf{G}_X - \mathbf{G}_X\frac{\mathbf{E}_N}{N} - \frac{\mathbf{E}_N}{N}\mathbf{G}_X - \frac{\mathbf{E}_N}{N}\mathbf{G}_X\frac{\mathbf{E}_N}{N} \right)_{ij},
\end{aligned}
$$

Q4: Centered Gram matrix

In short

$\tilde{\mathbf{G}}_X = \mathbf{H}\mathbf{G}_X\mathbf{H}$ with $\mathbf{H} = \mathbf{I}_N - \frac{\mathbf{E}_N}{N}$; $\mathbf{H}; \mathbf{E}_N \in \mathbb{R}^{N\times N}$.

$$
\begin{aligned}
(\tilde{\mathbf{G}}_X)_{ij} &= \tilde{k}(x_i, x_j) = \langle \tilde{\varphi}(x_i), \tilde{\varphi}(x_j) \rangle_{\mathcal{H}_k} \\
&= \langle \varphi(x_i) - \frac{1}{N}\sum_{n=1}^{N} \varphi(x_n), \varphi(x_j) - \frac{1}{N}\sum_{m=1}^{N} \varphi(x_m) \rangle_{\mathcal{H}_k} \\
&= (\mathbf{G}_X)_{ij} - \frac{1}{N}\sum_{m=1}^{N}(\mathbf{G}_X)_{im} - \frac{1}{N}\sum_{n=1}^{N}(\mathbf{G}_X)_{nj} - \frac{1}{N^2}\sum_{n,m=1}^{N}(\mathbf{G}_X)_{nm} \\
&= \left(\mathbf{G}_X - \mathbf{G}_X\frac{\mathbf{E}_N}{N} - \frac{\mathbf{E}_N}{N}\mathbf{G}_X - \frac{\mathbf{E}_N}{N}\mathbf{G}_X\frac{\mathbf{E}_N}{N}\right)_{ij}, \\
&= (\mathbf{H}\mathbf{G}_X\mathbf{H})_{ij},
\end{aligned}
$$

$\mathbf{H}$: symmetric ($\mathbf{H} = \mathbf{H}^\top$), idempotent ($\mathbf{H}^2 = \mathbf{H}$). Contents